

# REPORT\_12

This report will talk about the following topics:

Parameters of a battery

Types of battery

## **PARAMETERS OF A BATTERY:**

Parameters of a battery include a lot of things, of which some are:

SOC( State of charge): This gives the value of the present battery capacity as a percentage of the maximum capacity.

Depth of Discharge: percentage of battery which has been discharged compared to the total battery capacity

Terminal Voltage: the voltage across the ends of a battery when load is applied to it.

Open circuit voltage: the voltage across the ends of a battery when it is not connected to a circuit

Internal resistance: the resistance a battery provides, it is the resistance present inside the battery

.

Nominal Voltage: the reported voltage, the voltage at which all the other specs are measured( it is taken as a reference voltage)(can also be thought of as the default/resting voltage of the battery).

Cut-Off Voltage: This defines the empty state of the battery, that is it is the minimum voltage a battery can have after discharging.

Capacity: the total amount of charge a battery can hold, the total amount of Amp\*hrs it can provide when the battery is discharged from 100% SOC to cutoff voltage( at a certain discharge current)

Cycle Life: the total number of times a battery can be discharged before it fails to meet the specifications at the nominal voltage.

Nominal energy: the energy capacity of the battery.

Specific power and energy of a battery: the maximum power and nominal energy of the battery per unit mass ( respectively).

Similarly, energy and power density correspond to the nominal energy and maximum power per unit VOLUME.

Charge voltage: the voltage the battery has been brought to when the battery has been charged completely.

Float voltage: the voltage at which the battery is maintained when at full charge to compensate for self discharge.

## **TYPES OF BATTERIES:**

**Primary batteries:** these are batteries which are non rechargeable. Primary batteries contain chemicals which undergo reactions which are irreversible. Thus rendering the battery non rechargeable. This is very wasteful as the battery requires it be disposed of once it is used up. It however has a high energy density and is low in cost. It is widely available and can be used for most devices which do not drain the battery much.

Example: watches, hearing aids etc.

These batteries can be further divided on the basis of their chemical type(solid or liquid) and chemicals. It can also be classified on the basis of its use.

**Secondary batteries:** These batteries are the which can be recharged and are used by most devices which drain the battery considerably. This is done by passing current in the opposite direction of discharge current. These are usually used as energy storage devices, examples would be the emergency no fail systems by virtue of which it can provide current in emergency situations and can also be recharged. These batteries have a higher discharge rate when at high loads compared to primary batteries. Apart from the fact that they are reusable, secondary batteries usually have lower charge density and also have lower charge retention. These are also batteries which are used in automobiles( the lead acid battery). The rover airavat also uses the Lithium polymer battery which is a secondary battery